UNSW MARKETING ANALYTICS HACKATHON 2026 · FINALIST

# Getting slow loans funded faster

What 1.45 million Kiva loans say about predicting slow loans and supporting them







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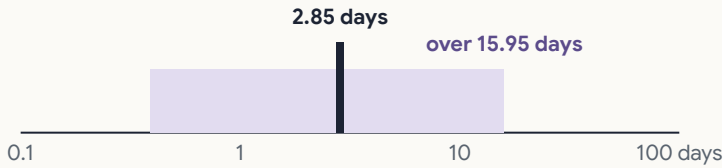
UNSW SYDNEY

# Act on predictable delays now; test writing advice before giving it

Problem, solution and proof across all evidence layers

THE PROBLEM

## 2.85 days

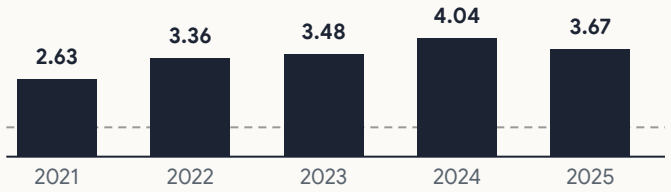



### Median wait to full funding

A quarter of loans wait over 15.95 days.

OUR SOLUTION

## 2.6–4.0×

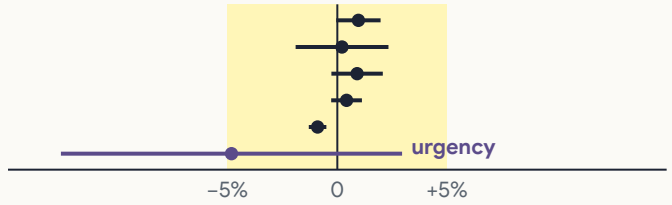



### Better than chance at flagging slow loans

Loan details flag the slowest fifth early, so support can go there first.

THE PROOF

## ±5%




### Framing links stay small

Five framings stay within ±5% of funding time, so rewriting stories is not the lever.

**Our ask to Kiva:** pilot a slow-funding flag with lender-side support, and test one truthful prompt.

Predict  Prioritise  Test

For whom and when: no period, loan size, gender or region shows a large framing effect; a few rare-word patterns are hypotheses to test.

# Agenda

01 → 02 → 03 → 04

## The problem

Who waits, and where?


How a loan gets funded 4
The wait 5
Where loans and delays sit 6


## What drives the wait

What does the evidence show?


What we read in each loan 7
Key insight 8
Four analyses 9
Country uncertainty 10
Programme controls 11
Why not rewrite stories 12
For whom and when 13
What text adds 14
Fragile across countries 15


## Our solution

What can Kiva do now?


Predict, prioritise, test 16
Predict 17
Prioritise 18
Test 19

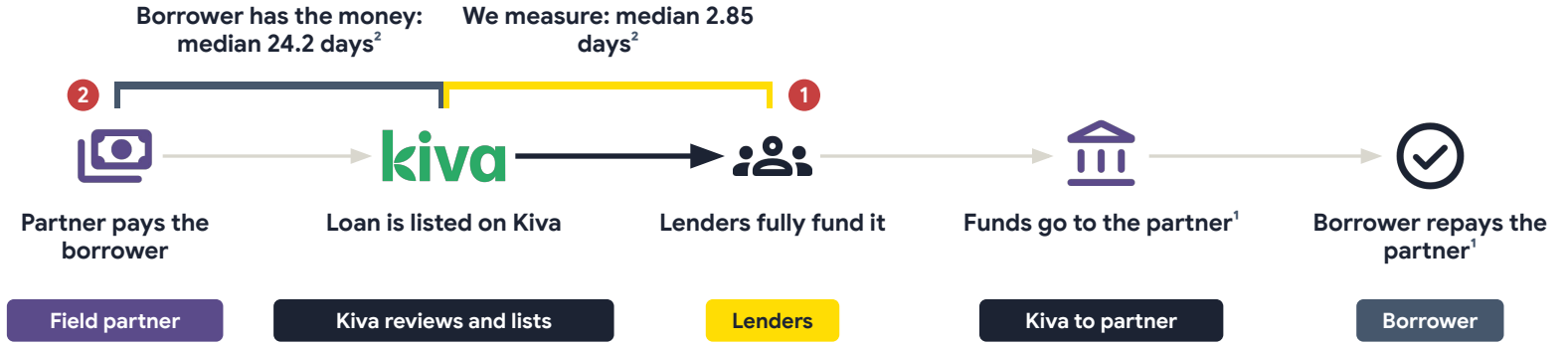

## Safe next steps

What keeps it honest, and what comes next?


Guardrails 20
Roadmap 21

# Borrowers usually have the money before lenders fund the loan

Descriptive: how a typical pre-disbursed Kiva loan moves, who acts at each stage, and what we measure




### Loans disbursed to the borrower before listing




### Who is in the data

**In the data:** 1,453,840 completed loans, including 1,637 refunded

**Not in the data:** listings that expired or never funded 4

**Figure 1.** Journey of a typical pre-disbursed Kiva loan, not to scale; medians from the full-data audit.

[1] Typical field-partner flow; these steps are not observed in the data.

[2] Separate medians: 24.2 days covers the 96.4% disbursed before listing, and the two do not add up along one loan.

**Scope:** 1,453,846 input records; 6 negative durations excluded; 1,637 refunded records included.

Independent hackathon analysis; not affiliated with or endorsed by Kiva.

### 1 We study the lender-side step

Our outcome runs from **listing to full funding**, median 2.85 days.

### 2 Borrowers usually already have the money

**96.4% of loans** reach the borrower a median **24.2 days** before listing.

### 3 Speed turns over partner capital

Only **3.6%** are paid out after listing, so slow funding mainly holds up the partner's next loans.

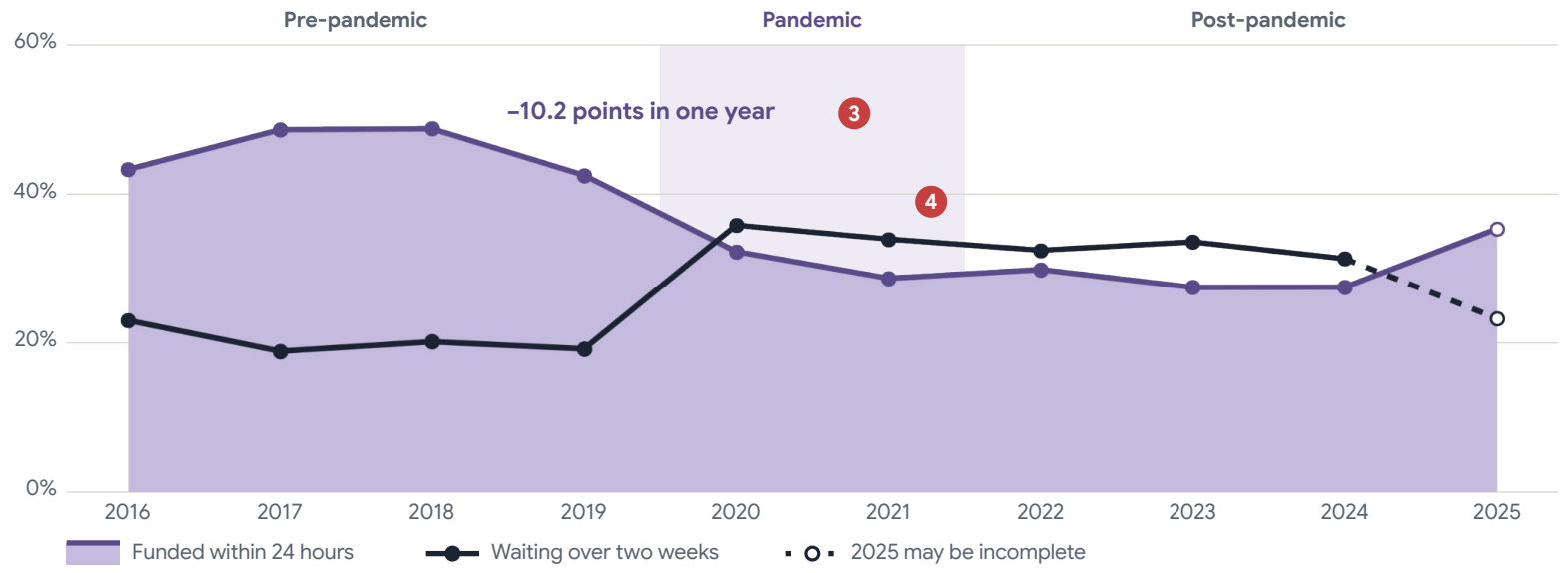
### 4 Unfunded loans are missing

Only loans that **eventually funded** are in the data; expired listings are not.

→ Why speed is a lender-side problem

# Waits got longer after 2019: a third of loans now wait over two weeks

Descriptive: platform funding time from listing to full funding, 1,453,840 completed loans, by posting year



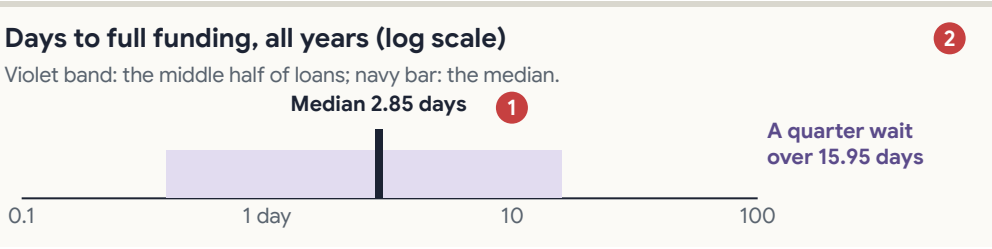






**Figure 2.** Same-day share and share waiting over two weeks by posting year; funding-time quartiles, all years.

### 1 Most loans fund within days

Half are fully funded within **2.85 days**, a quarter within **0.39 days**.

### 2 A long tail waits weeks

A quarter of loans wait **over 15.95 days**, while borrowers already have the money.

### 3 Same-day funding fell in one year

From **42.5% in 2019 to 32.3% in 2020**, a fall of 10.2 points.

### 4 Long waits became common

Loans waiting over two weeks rose from **19.2% to 35.8%**; in 2021 that was **55,408 loans**.

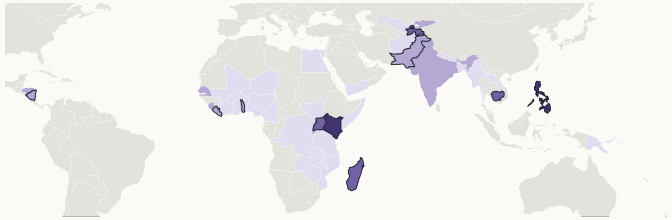
→ Why early warning matters

# Over half of all two-week waits come from just three countries

Descriptive: where loans sit and where they wait over two weeks, by country, 1,453,840 completed loans

### (a) Where the loans are

Ten of 48 countries hold 81% of loans (outlined on the map); table share cells use the map's colour scale.




| | Country | Region | Loans | Share | Same-day |
|---|---|---|---|---|---|
| 1 | **Philippines** | Asia | 521,586 | 35.9% | 48% |
| 2 | **Kenya** | Africa | 266,562 | 18.3% | 29% |
| 3 | **Uganda** | Africa | 69,651 | 4.8% | 31% |
| 4 | **Tajikistan** | Asia | 68,865 | 4.7% | 18% |
| 5 | **Cambodia** | Asia | 58,074 | 4.0% | 49% |
| 6 | **Madagascar** | Africa | 55,670 | 3.8% | 67% |
| 7 | **Pakistan** | Asia | 42,767 | 2.9% | 15% |
| 8 | **Nicaragua** | Central America | 38,520 | 2.6% | 15% |
| 9 | **Liberia** | Africa | 28,847 | 2.0% | 28% |
| 10 | **Togo** | Africa | 27,513 | 1.9% | 38% |

### (b) Where the long waits are

Each bar is all of a country's loans, split by how long they took to fund.

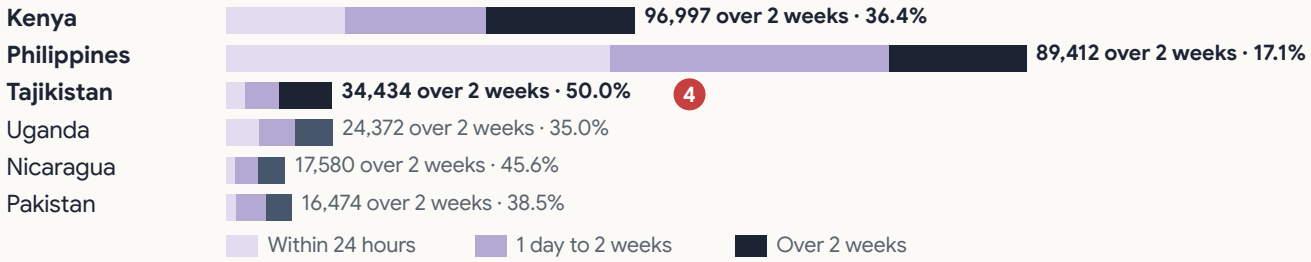



**Figure 3.** Share of all loans by country (a, map and table) and loans waiting over two weeks (b); same-day: funded within 24 hours.

### 1 Ten countries hold most loans

Ten of 48 countries hold **81% of loans**; Asia and Africa hold **92.8%**.

### 2 Speed varies widely by country

Same-day funding runs from **15% in Pakistan to 67% in Madagascar**.

### 3 Three countries hold half the long waits

Kenya, the Philippines and Tajikistan hold **56.1%** of the **393,847** loans that waited over two weeks.

### 4 Large markets wait too

In Kenya **36.4%** of loans wait over two weeks; in Tajikistan **50.0%**.

→ Why the flag must work country by country

# Each loan story is scored on eight framing measures

Illustrative listing: what is fixed when a loan is listed, and what we measure in the description



**FIXED AT LISTING** 1

| Amount, median | Term, median | Borrowers, median |
|---|---|---|
| **US$325** | **12 months** | **1** |
| Sector, most common | Country, most common | Partner |
| **Agriculture** | **Philippines** | **Field partner** |

**EDITABLE DESCRIPTION** 4

> She has a banana stall at the market to support her family. She wants to buy more stock quickly, before prices rise. She is thankful to every lender who helps her business grow.

**The eight framing measures, counted per 100 words, with the share of loans using each (plus a sentiment score)** 2

- **Family** — Relatives and schooling — e.g. children · in 88.7%
- **Business** — Trade and stock words — e.g. market · in 85.0%
- **Basic needs** — Water, health, housing — e.g. water · in 23.3%
- **Agency** — Planning and self-direction — e.g. decided · in 54.1%
- **Gratitude** — Thanks and blessings — e.g. thankful · in 19.6%
- **Urgency** — Time-pressure words — e.g. quickly · in 1.3% 3
- **First person** — The borrower's own voice — e.g. my · in 1.0%
- **Third person** — Written about the borrower — e.g. her · in 97.3%

**Figure 4.** Illustrative composite listing, not a real borrower; dataset medians and most common categories. Word lists not yet validated (gate 1; methods A7).

### 1 Most of the listing is fixed

Amount, term, sector, country and partner are set; the **description** is the part a partner writes.

### 2 We score eight framing measures

Counted with **word lists per 100 words**, plus VADER sentiment and text form.

### 3 Some measures are rare

Urgency appears in **1.3%** of descriptions and first person in **1.0%**; only **11.3%** never mention family.

### 4 Pronouns mostly reflect the writer

Descriptions are **written about the borrower**, so third person is context.

→ Why wording is tested, not assumed

# Slow funding is a lender-side delay that loan details predict, not wording

Three findings across the outcome audit, the early-warning check and the equivalence test, each leading to a step of the solution

### The wait is on the lender side

1

**96.4%**

of loans reach the borrower before listing




→4

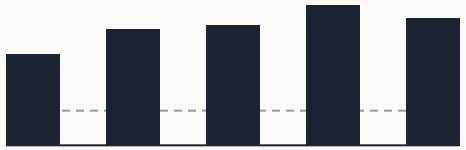
Leads to **Prioritise**

### Loan details predict who waits

2

**2.6–4.0×**

better than chance at flagging the slowest fifth




→17

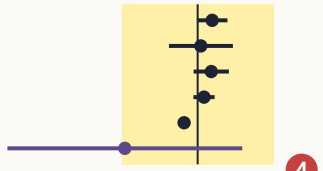
Leads to **Predict**

### Wording makes little difference

3

**±5%**

five framings stay within this share of funding time




→12 →13

Leads to **Test first**

## 1 Act on the lender side

**96.4%** of loans reach the borrower before listing, so the wait ties up partner capital, not the borrower.

## 2 Predict slow loans early

A boosted tree on loan structure flags the slowest fifth **2.6–4.0× better than chance**.

## 3 Don't chase framing tweaks

Five framings stay **within ±5%** of funding time, and text adds little to the flag.

## 4 No group stands out

**No group** shows a confirmed large framing effect; family and third person stay small in **all 10 subgroups**.

→ The evidence behind each insight follows

# Four analyses feed the solution; none is a causal claim

One dataset: the equivalence test builds on the robustness checks, and the forecast benchmark asks a separate question

**Kiva loans, 2016–2025: 1,453,840 valid completed durations in 48 countries** 1

### Explanatory models

**Which features are linked to funding speed?**



64 of 120 significant terms lost significance

Feeds Test 2

### Robustness checks

**Do the links survive richer controls?**



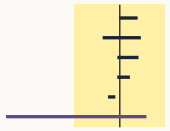
1 of 36 passes

One tiny link passes, in one model only

Feeds Prioritise 3

### Equivalence test

**Which links are small enough to rule out a large effect?**



Five framings small; urgency unresolved

Feeds Prioritise

### Forecast benchmark

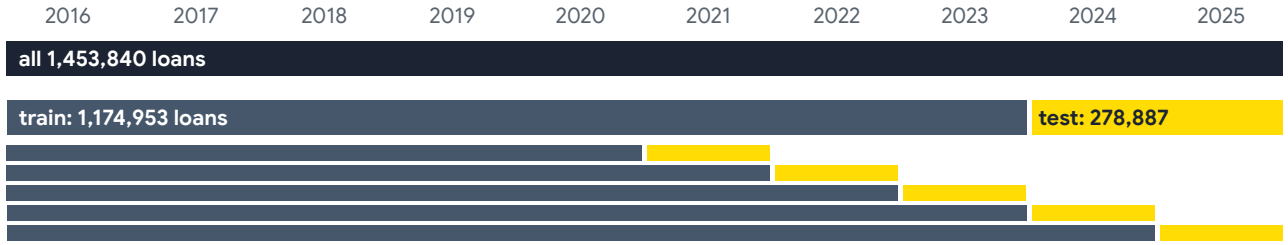
**Does text improve a forecast built on loan structure?**



Small gain: 0.26–0.39 days, about half context

Feeds Predict 4

**Years each analysis uses**




Details: forecast cutoffs A2; primary model scores A3; text methods A7.

### 1 One dataset, four questions

All four use the same **1,453,840 loans**, and each supports a different kind of claim.

### 2 Uncertainty comes first

Clustering by 48 countries removed **64 of 120** significant terms.

### 3 Framing signals stay small

After a wild cluster bootstrap, **1 of 36** tests passes: third person, in one model only.

### 4 Forecasting is tested separately

Each model trains only on **loans completed before its test year**.

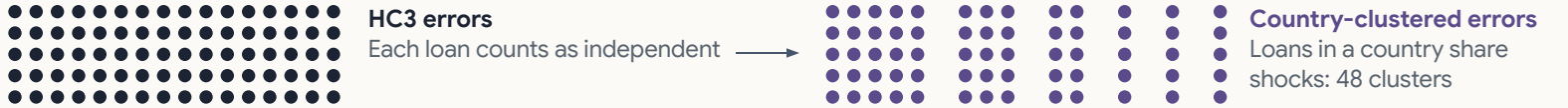
→ None is causal, so the solution ends with a test

# Clustering by country removed 64 of 120 significant terms

Explanatory models: HC3 versus 48-country clustered errors on 1,453,840 loans

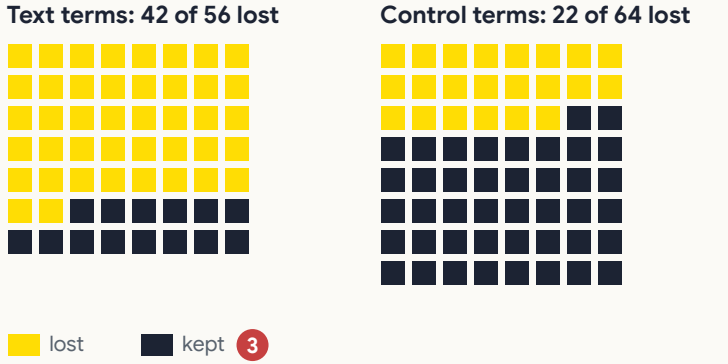
### Why the errors widen

The loans and estimates stay the same; only the assumption about how loans vary together changes.




### Which terms lost significance

Each square is one term significant under HC3.




### Urgency before and after

p-value for urgency in each model; clustering moves both past the 0.05 line.

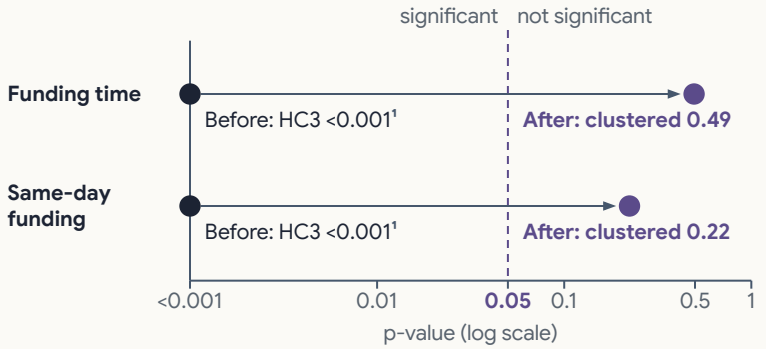



**Figure 5.** Explanatory models, HC3 versus errors clustered by country (48 clusters). Dots in the top panel are illustrative. Regional check: A1.

[1] Printed as 0.0000 in the saved run. Text terms include interactions of a text measure with a control.

### 1 Few countries dominate

Ten countries hold **81% of loans**; regional patterns rest on **two countries each**.

### 2 Text terms are hit hardest

**42 of 56 text terms** lost significance, against 22 of 64 controls.

### 3 Few text terms survive

Only **14** stay significant: sentiment in both models and 12 family-framing interactions.

### 4 Urgency no longer passes

Its p-value rose from below 0.001 to **0.49** for funding time and 0.22 for same-day funding.

→ Why the flag is checked country by country

# One tiny signal passes the strictest test, but not with programme controls

Robustness checks: programme controls, then a wild cluster bootstrap across 48 countries

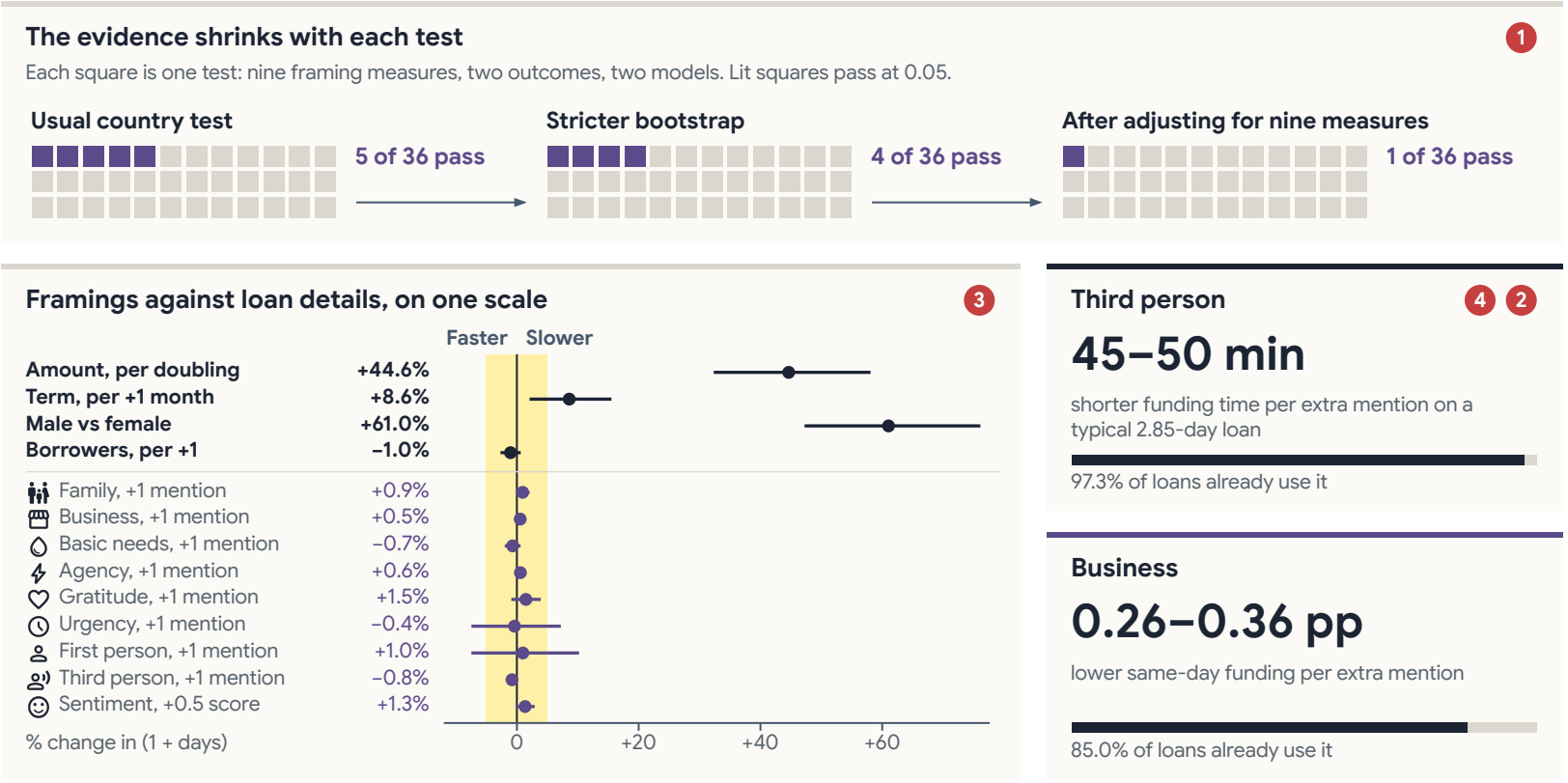



**Figure 6.** Per extra mention per 100 words; BH-adjusted wild cluster bootstrap p across nine terms (Webb weights, 9,999 draws). Typical loan: the 2.85-day median. **Scale.** Change in (1 + days), not days, with 95% country-clustered ranges; programme-adjusted model. Amount: a doubling from the US$325 median, within its size band.

### 1 The evidence shrinks with each test

**5 of 36** tests pass the usual country test, 4 the bootstrap, **1** after adjustment.

### 2 The one that passes is tiny

Third person, expanded model: **−0.9%** per mention (BH p 0.008); with programme controls, BH p 0.426.

### 3 Loan details are on another scale

A doubled amount goes with **+44.6%** and a month of term **+8.6%** in (1 + days); framings stay near ±1.5%.

### 4 Pronouns may reflect the writer

Third person is in **97.3%** of loans, stays within ±2%, and partners usually write the text.

→ Why we prioritise loans, not wording

# Five framings have small links to funding time; urgency remains unknown

Equivalence test of the robustness-check coefficients; exploratory, bound set after the estimates

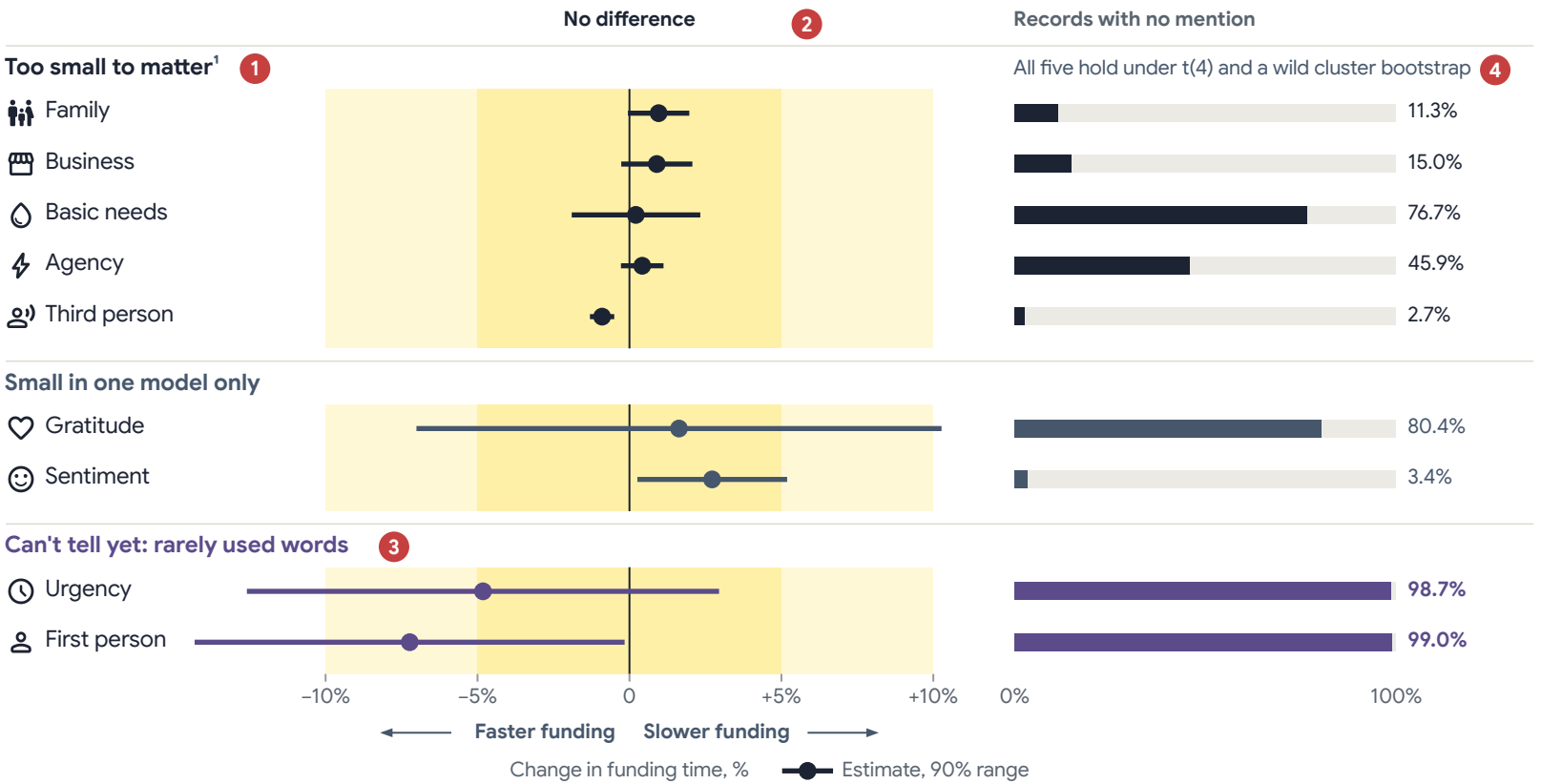



**Figure 7.** Change in funding time per extra mention per 100 words (sentiment: per +0.5 score), expanded model, 1,453,840 loans.

[1] Within ±5% of funding time: ±0.05 on log(1 + days), BH-adjusted; bound set after the estimates.

### 1 Five framings have small links

Family, business, basic needs, agency and third person stay **within ±5%** in both models.

### 2 Small means under five hours

±5% is about **4.5–4.7 hours** on a typical 2.85-day loan.

### 3 Urgency is too rare to judge

Under **2% of descriptions** use urgency or first person; their ranges still reach **6–14%**.

### 4 The verdicts hold under stricter tests

A wild cluster bootstrap keeps **35 of 36 verdicts**; three framings stay small even within **±2%**.

→ Why we don't chase framing tweaks

# Family and third person stay small in every group; most others unconfirmed

Exploratory: programme-adjusted model refitted in 10 subgroups, wild cluster bootstrap, same ±5% bound as the last slide

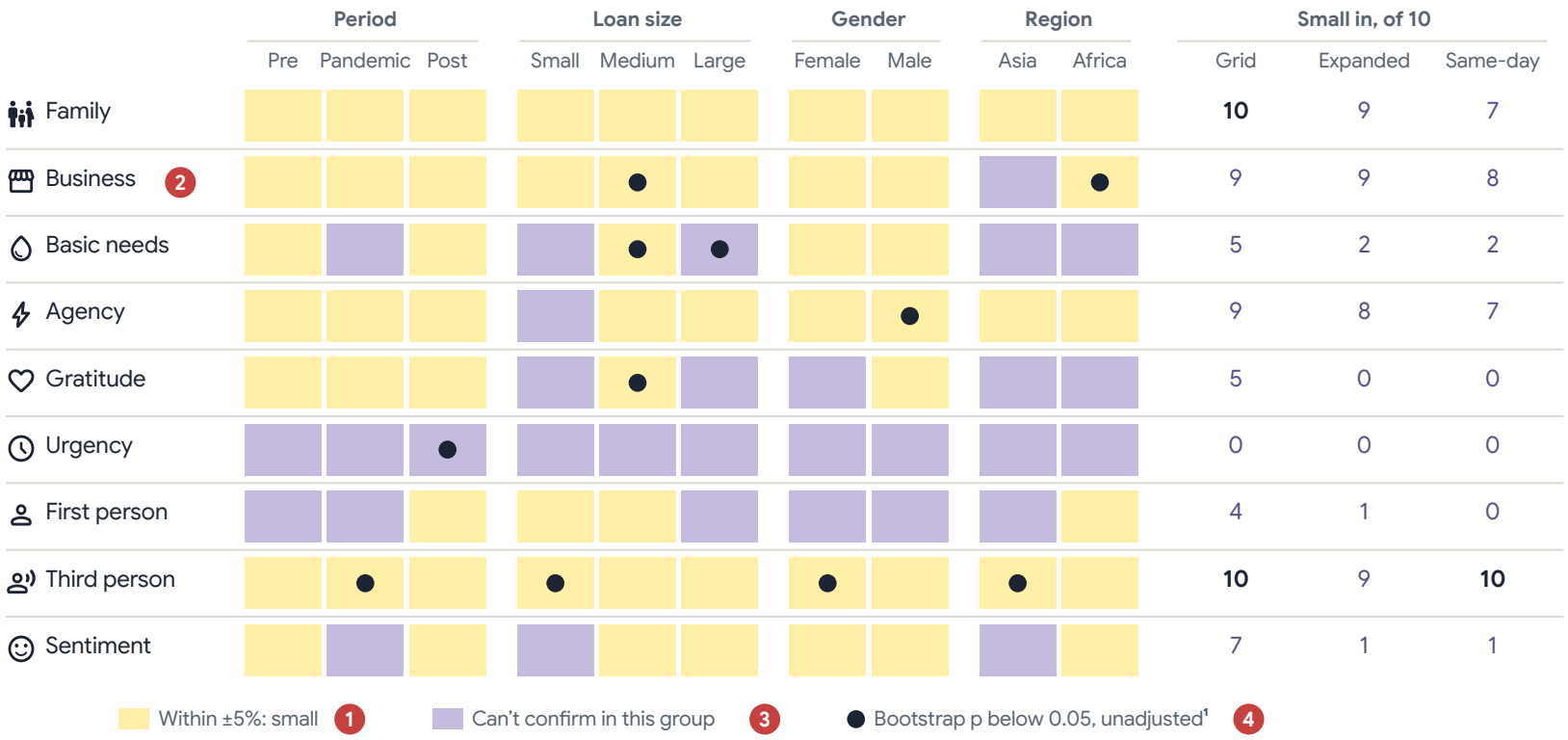



**Figure 8.** Funding-time verdict by framing and subgroup, programme-adjusted model, wild cluster bootstrap, BH within each subgroup. Expanded: funding time. Same-day: ±1 pp.
[1] Of these 90 cells; none passes BH. Hypotheses for the Test step, not findings.

### 1 Two framings hold in every group

**Family** and **third person** stay within ±5% in **all 10 subgroups**.

### 2 Two more miss one group

**Business** misses only in Asia (BH p 0.052); **agency** only in small loans (0.061).

### 3 The rest can't be confirmed

Basic needs, gratitude and sentiment hold in **5, 5 and 7 of 10**: unconfirmed, not large.

### 4 No confirmed subgroup signal

**6 of 360** tests pass BH, none of them **urgency or agency**.

→ Why: no group shows a confirmed large framing effect

# Text adds a small, inconsistent forecast gain; about half is context

Forecast benchmark: gain from adding text to loan details, five annual test years, identical rows

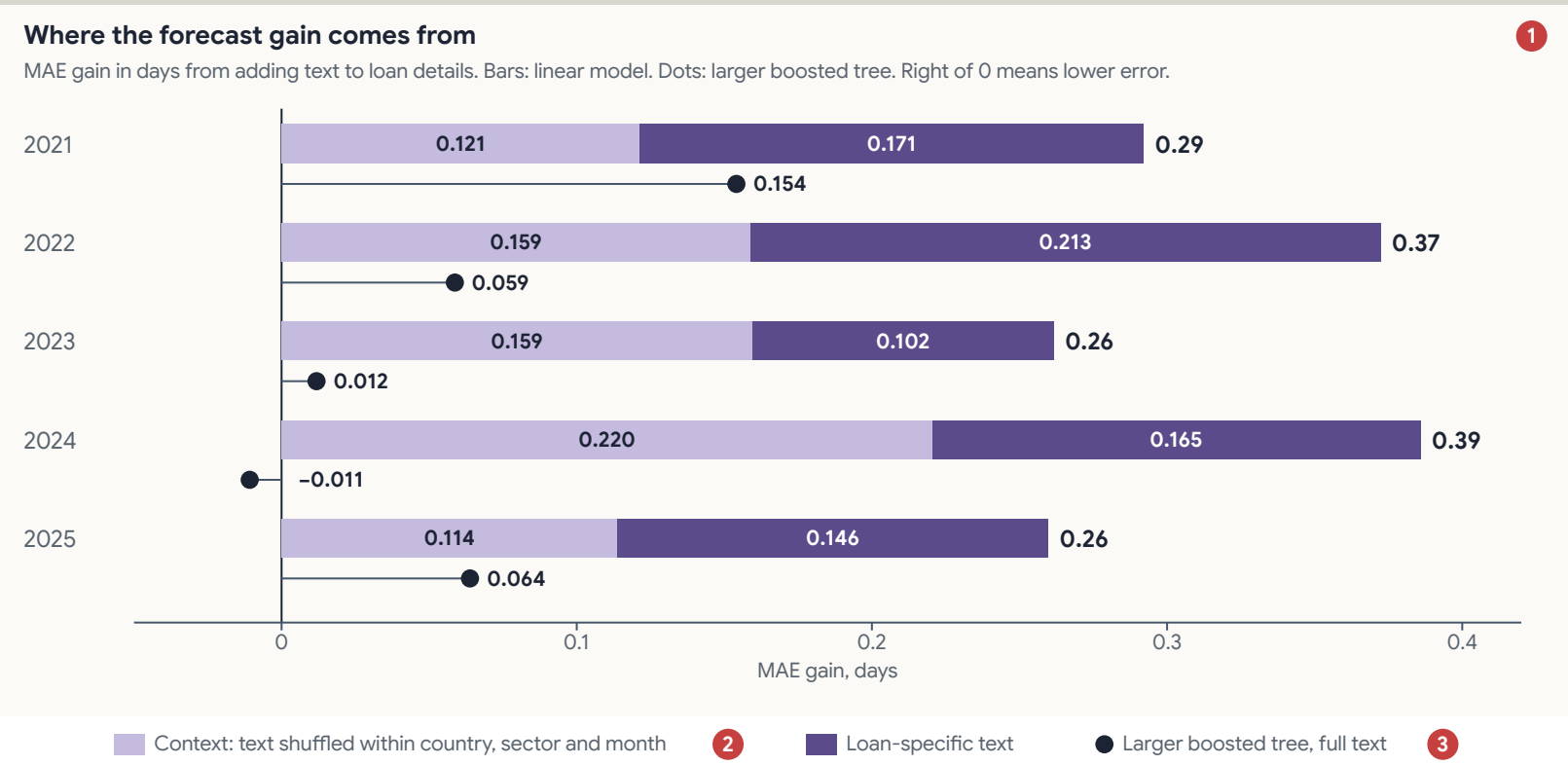



**Figure 9.** Annual MAE gain in days, identical rows; bars from the placebo refit, within 0.003 days of the benchmark. Text methods: A7. Stage split: A6.

### 1 A small gain on a simple model

Text, mainly **TF-IDF word patterns**, lowers a linear model's error by 0.26–0.39 days, about 7 hours.

### 2 About half is context

Shuffled text keeps **42–61%** of the gain; the loan-specific part is 0.10–0.21 days.

### 3 A stronger tree gains less

Fixed trees gain up to **about 0.15 days** from text; tuned trees reach **0.16–0.22** in 2021 (A3).

### 4 It does not show rewriting helps

Text alone does worse than loan details, and no forecast tests an edit.

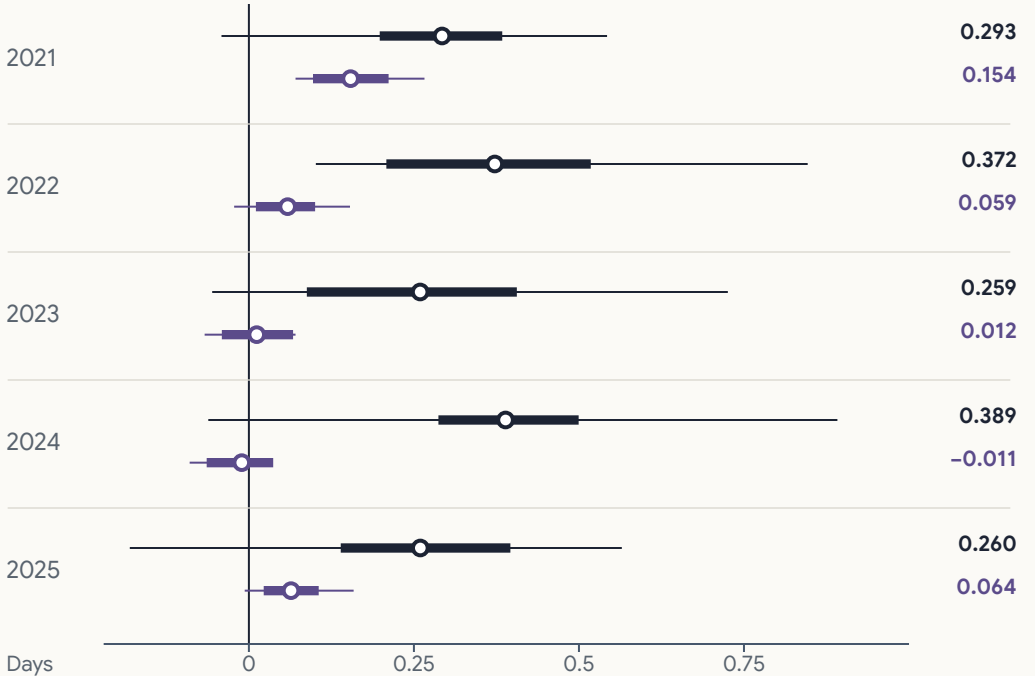
→ Why we predict with loan details first

# The text gain is fragile across countries and on a stronger model

Forecast benchmark: annual MAE gain in days, with 95% ranges from resampling months or countries

### The text gain by year: two models, two ranges

MAE gain in days from adding text to loan details; right of 0 means lower error. The dot is the gain.




Linear model | Thick: month range (1) | Larger boosted tree (2) | Thin: country range (3)

### How often the gain clears zero

Years out of 5 where the 95% range excludes 0. More years means a steadier gain.

By month | By country

**Linear model**
Holds over time, not across countries

5 ●●●●● → 1 ○●○○○

**Larger boosted tree**
Less consistent even over time

3 ●●○○● → 1 ●○○○○

**Loan-specific part of the text gain**
The loan's own words; still uneven

4 ●●○●● → 2 ○●○○●

● Range excludes zero ○ Includes zero

**Figure 10.** Annual MAE gain in days from adding text, identical rows. Text = form, word-list counts, VADER sentiment and TF-IDF word patterns. Details: A6, A7.

[1] Forecasts capped at the longest training wait (173.4 days); a diagnostic, not the reported model. (4)

### 1 Stable across months, linear only

The linear gain clears zero **every year** when months are resampled.

### 2 A stronger tree gains little

Fixed trees gain up to **about 0.15 days**, tuned trees 0.16–0.22 in 2021; the linear model gains 0.26–0.39.

### 3 Across countries it is fragile

**Few years clear zero** once countries are resampled, even for the loan-specific part.

### 4 Extreme forecasts explain part of it

With forecasts capped[1], the linear gain's country range includes zero in only **1 of 5** years.

→ Why the flag needs validating by country

# Our solution: predict slow loans, prioritise support, test before advising

A three-step framework for Kiva and partners, built only on what the evidence supports

## 1 Predict

**What**
**Score each listing's risk of slow funding when it is posted.**

**Who**
kiva platform

**When**
At posting

**Status**
Retrospective check only

Evidence: Slide 17

## 2 Prioritise

**What**
**Send lender-side support to flagged loans.**

**Who**
kiva and partners

**When**
While the loan raises funds

**Status**
Needs a pilot

Evidence: Slides 12–13 and 18

## 3 Test

**What**
**Earn the right to give writing advice.**

**Who**
Research with partners

**When**
Before any writing advice

**Status**
Not started

Evidence: Slide 19

Independent hackathon analysis; not affiliated with or endorsed by Kiva.

### 1 Start with what is known at posting

A listing's risk can be scored the moment it is posted, **from loan details**.

### 2 Act on loans, not wording

Support goes to flagged loans, because **most framing links stay small**.

### 3 Advice comes last

No description edit has been tested, so **writing advice waits for a pilot**.

### 4 Nothing is deployed yet

The flag is checked on past loans only, and the **support levers still need a controlled pilot**.

→ Why we start by flagging slow loans

# A flag at listing beats a simple historical lookup in every year

Concept to pilot: retrospective check on completed loans only, five annual test years

Concept to pilot

4

Listing at posting time

**Agriculture · US$325 · 12-month term**

Illustrative listing, not a real borrower

Predicted risk of slow funding

top 10%

**Flag: likely slow to fund**

Chance of same-day funding: low

**Suggested support**

- Feature it to lenders
- Offer matching funds
- Alert lenders who follow the sector
- Ask the partner to review it

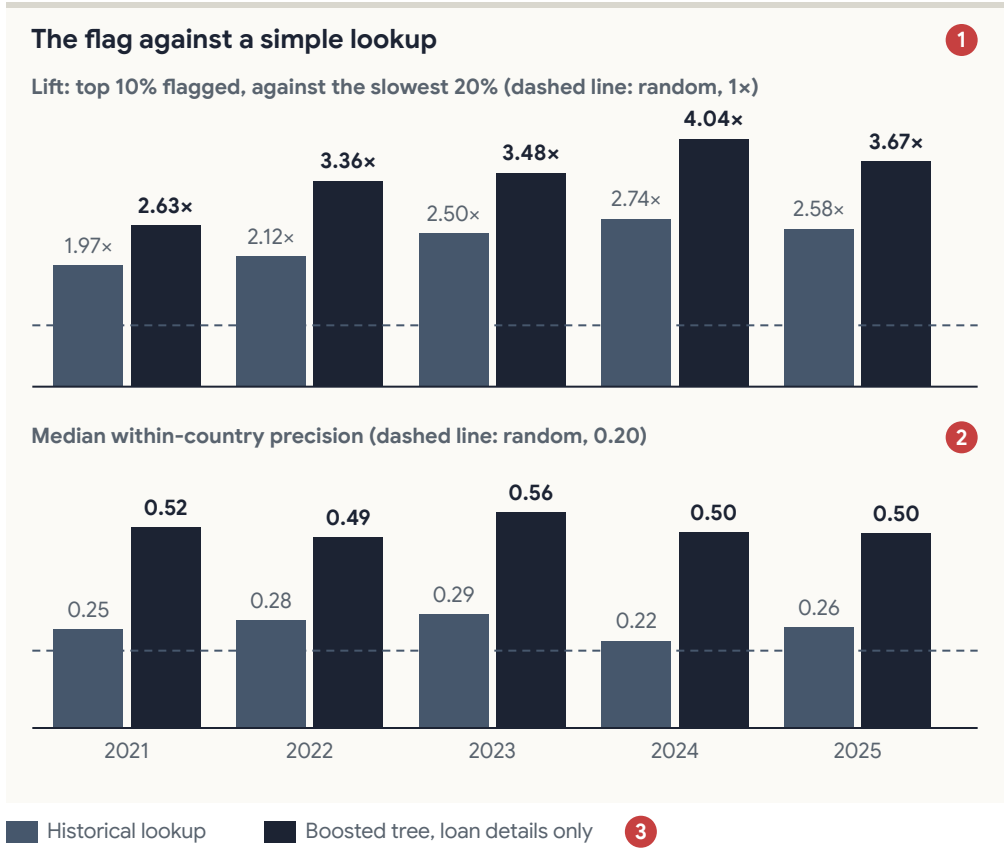



**Figure 11.** Top 10% of predicted-slow loans against each test year's slowest 20%. Lookup: training-only median by country, sector and month. Model scores: A3.

### 1 Better than a simple lookup

Flagging the top 10% gives **2.63–4.04×** lift, against 1.97–2.74× for a lookup; the gap's 95% range is **above zero every year**.

### 2 Clearest within a country

Median precision is **0.49–0.56**, against 0.22–0.29 for the lookup; random is 0.20.

### 3 Text adds little to the flag

Adding text changes the tree's precision by **−0.003 to +0.005**.

### 4 Checked looking back only

Completed loans only; **precision ranges 0.18–0.90** across the largest countries.

→ Why flagged loans come first

# Send lender-side support to flagged loans, not story edits

Proposed levers for Kiva and partners; their effect on speed is to be tested, not yet measured

### Feature it 1

Show flagged loans to lenders early on the lending page.

**Who:** kiva

**Watch:** crowding out other loans 2

### Match it

Offer matching funds where a loan is likely to stall.

**Who:** kiva and funders

**Watch:** the cost of matching funds

### Alert lenders

Notify lenders who follow the loan's sector or country.

**Who:** kiva

**Watch:** lender fatigue

### Follow up

Ask the field partner to check the listing is complete and clear.

**Who:** Field partners

**Watch:** extra partner workload

### Don't chase framing tweaks 4

The evidence points away from wording, so effort goes to the loan.

**Instead**

- Flag slow loans at posting
- Reach the right lenders
- Pilot before any advice





**Scope: the levers are proposals; our data do not measure their effect.** 3

Independent hackathon analysis; not affiliated with or endorsed by Kiva.

### 1 Four levers act on lenders

Each changes **who sees a flagged loan** or what support it gets, not the story.

### 2 Each has a cost to watch

Featuring can crowd out other loans; matching uses money; alerts can **tire lenders**.

### 3 Their effect is unmeasured

Our data cannot say what these levers do, so **each needs a controlled pilot**.

### 4 Wording is not the lever

The evidence so far gives **no reason to rewrite stories** to speed funding.

→ Why writing advice must pass gates first

# Writing advice has to pass three gates first

Proposed next research for Kiva and participating partners; each gate must pass before the next

### 1 Validate the measures

**Question** 1
**Do our word counts capture clarity, need and agency?**

**Do**
Build a **small codebook**; **two people** code the same narratives and resolve disagreements.

**Done when**
Definitions are **frozen** before later periods are checked.

### 2 Close the data gaps

**Question** 2
**What did lenders see, and which listings never funded?**

**Do**
Get **original texts**, partner identifiers, exposure data and **expired listings**.

**Done when**
**Unfunded listings** and **listing-time text** are in the data.

### 3 Pilot one prompt

**Question** 3
**Does editing a description change funding speed?**

**Do**
With partners, test **one truthful clarity prompt** against a **control**; urgency in Asia is only a **weak signal** (p 0.107).

**Done when**
Threshold, power and guardrails are **agreed in advance**.

**Each gate unlocks a stronger claim**

**Today**
Associations and forecast information

1 **After gate 1**
Measures that mean what we say

2 **After gate 2**
Results for every listing, not only funded ones

3 **After gate 3**
A tested effect of editing

**If a gate fails, stop there: report the uncertainty and give no writing advice.** 4

## 1 Check the measures first

Word counts may not capture clarity, need or agency until **people have coded them**.

## 2 Then fill what we cannot see

We lack **what lenders saw** and the listings that never funded.

## 3 Only a pilot can test an edit

One truthful prompt against a control, with **the pass mark set in advance**.

## 4 A failed gate stops the work

The claim stays where it is, and **no writing advice goes out**.

→ Why the limits stay on the page

# Four limits decide what the solution can claim today

What this evidence cannot show, and what to do about it

| Limit | What we cannot show | What we do about it |
|---|---|---|
| **Associations, not causes** 1 — Limit | Every result is **observational**; **no edit or lever** has been tested. | **Pilot** each lever and prompt against a **control**. |
| **Funded loans only** 2 — Limit | Listings that **expired unfunded** are not in the data, so **reach** is unknown. | Add **expired listings** before judging reach. |
| **Uneven across countries** 3 — Check | Flag precision **varies widely** within the largest countries, and probabilities **drift** in 2025. | **Recalibrate and validate** the flag country by country. |
| **For whom and when** 4 — Checked | **No group** shows a confirmed large effect, and the one strict-test pass holds in **one model only**. | Treat **weak signals** as hypotheses to test. |

**Where each limit gets addressed**

**Flag rollout** validated by country 3 → **Gate 1** measures checked → **Gate 2** missing data added 2 → **Gate 3** one pilot run 1 4

### 1 Claims stay at association level

We say **linked with**, not caused by, until a pilot has run.

### 2 Reach stays an open question

We cannot yet say a lever helps **loans that would otherwise expire**.

### 3 The flag is validated by country

No country gets a flag **until it is recalibrated** on local data.

### 4 One passing signal is not a rule

It holds in one model only, so **no writing rule follows**.

→ Why the roadmap starts with a pilot

# Start with prediction and support now; earn writing advice later

Proposed roadmap for Kiva and partners

## Now 1

**Predict**

**Pilot the flag** at listing

**Prioritise**

Route **lender-side support** to flagged loans

**Prioritise**

Stop investing in **framing tweaks**

## Next

**Predict** 2

**Recalibrate and validate** the flag by country

**Test · Gate 1**

Validate the **text measures** with a codebook

**Test · Gate 2** 3

**Data partnership**: listing-time text, expired listings, partner identifiers

## Later

**Test · Gate 3**

Pilot **one truthful prompt** against a control

**Test**

Answer **for whom and when** properly

**Test** 4

Give **writing advice** only if the pilot shows it helps

### Our ask

**One partner** to pilot the flag

**Expired listings** and **country data**

A partner for **one prompt pilot**

### 1 Start where no rewrite is needed

The flag and lender support can be piloted **without changing a story**.

### 2 Prove the flag before scaling

Country checks come **before any wider use**.

### 3 Data comes before a writing test

Measures and missing data come **before a prompt is piloted**.

### 4 Writing advice is the last step

It follows **only if the pilot shows it helps**.

→ Why our ask starts with the flag

Thank you

# Flag slow loans first. Earn writing advice last.

**Mike Nguyen and Sophia Nguyen**
Team Cultural Blend - University of Technology Sydney
Code and evidence: github.com/tuannm3812/unsw-ma-hackathon-2026



# Regional family links look strong but rest on two countries each

Explanatory models: within-region slopes with country-clustered and heuristic few-cluster intervals

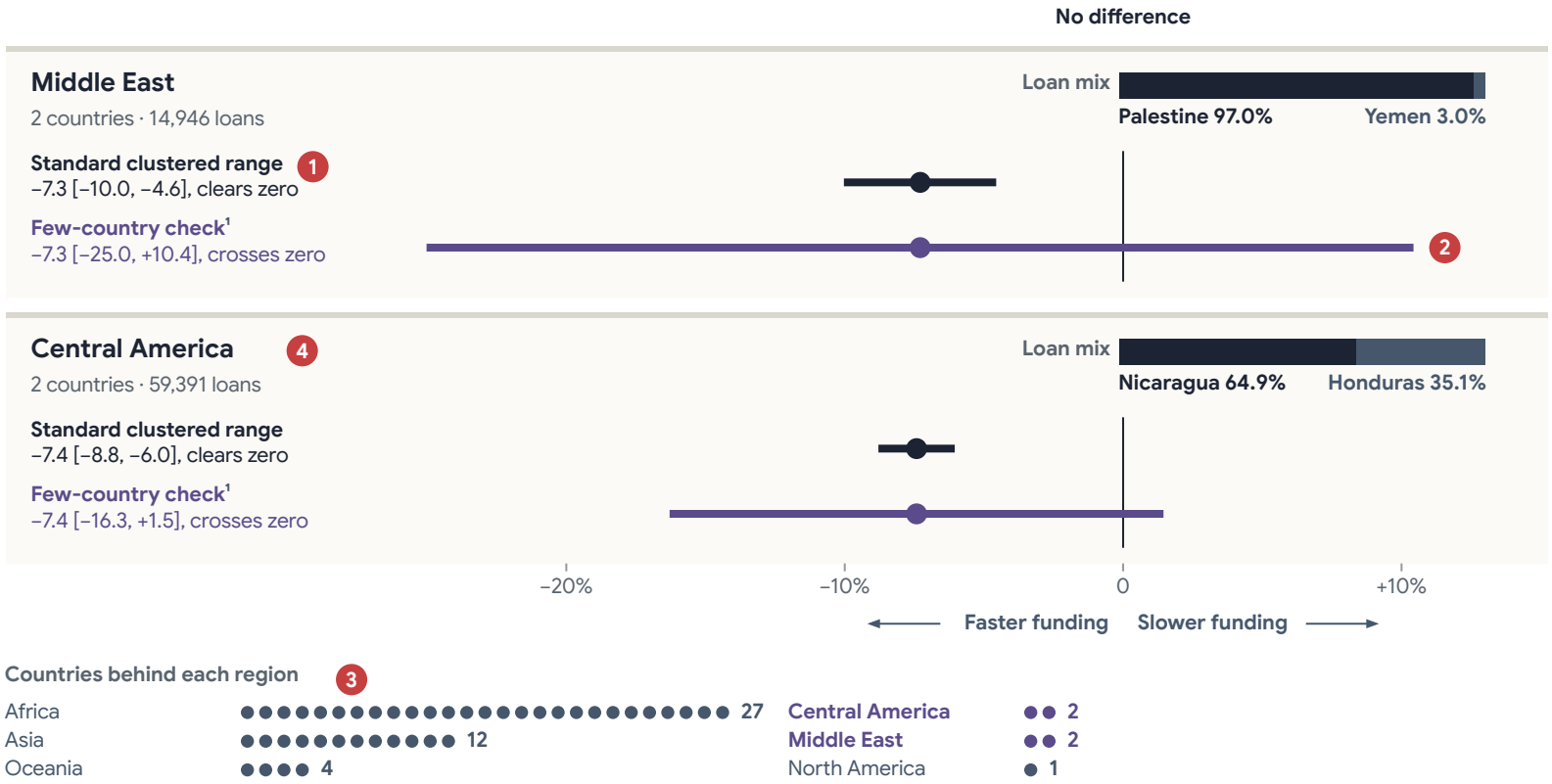



**Figure A1.** Within-region family-framing slopes, 95% ranges, explanatory models; one dot per country, Kiva regions.

[1] t(1) reference on the same clustered error: a heuristic screen, not a calibrated test. Axis in log points × 100.

### 1 Both links look strong at first

Standard errors put family framing at about **7% faster funding** per extra mention in each region.

### 2 Two countries cannot carry them

A few-country check widens both ranges **across zero**; the Middle East is **97% Palestine**.

### 3 Other regions have many more countries

Africa has **27 countries** and Asia **12**; these two regions have two each.

### 4 No regional writing rules yet

Treat both as **descriptive patterns**; a real test needs more countries or partner-level data.

# Each forecast trains only on loans completed before its cutoff

Forecast benchmark: training records must have completed before each cutoff

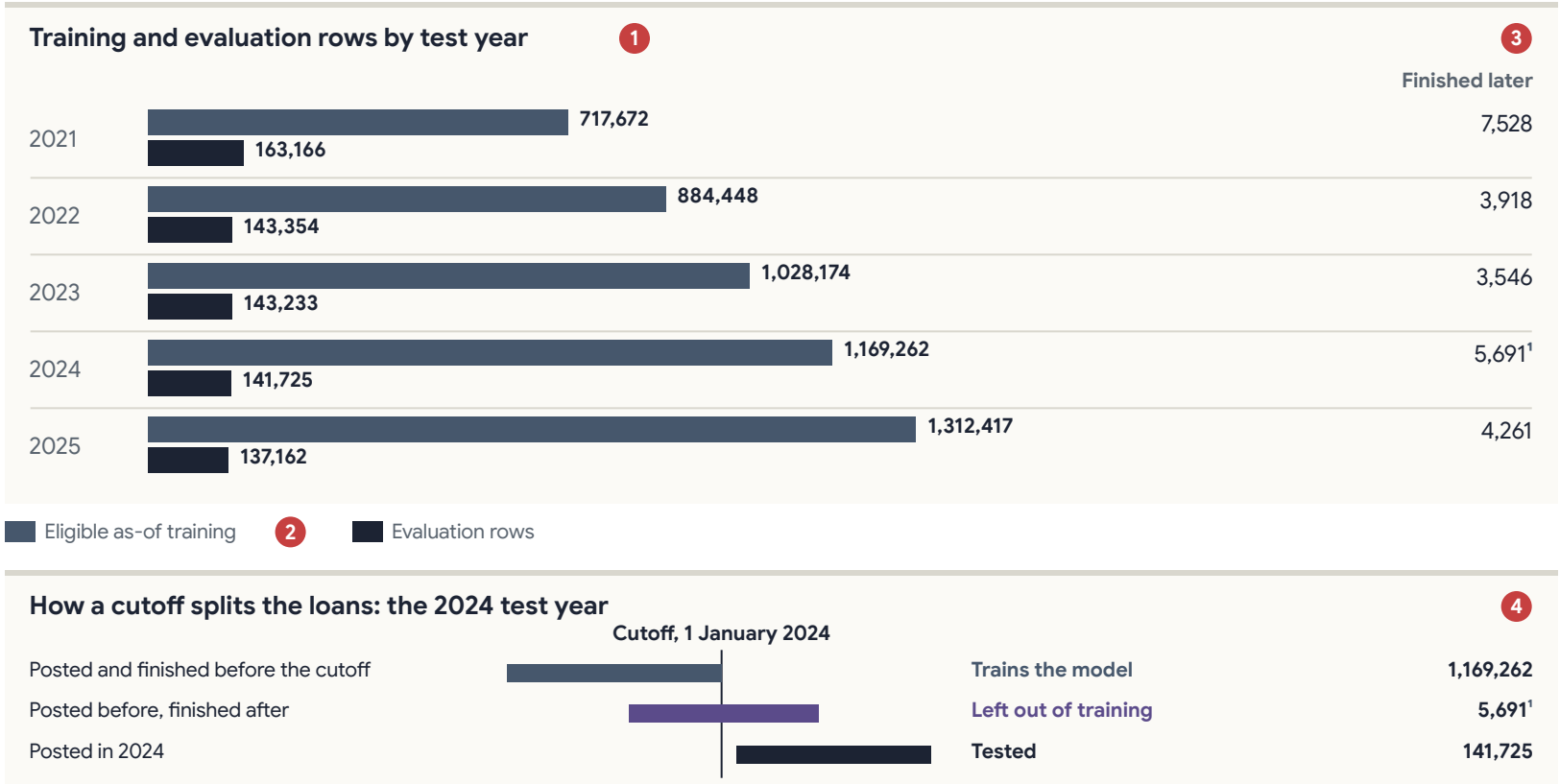



**Figure A2.** Training availability and evaluation rows by year, with the 2024 split as a worked example.

[1] A posting-only split would keep these 5,691 training records; evaluation windows still select eventual completions.

### 1 Training looks back only

A record counts only if **funding finished before the cutoff**.

### 2 The training set grows each year

From **717,672** loans in 2021 to **1,312,417** in 2025.

### 3 Late finishers are left out

Loans posted before a cutoff but **finished later** do not train that year's model.

### 4 2024 shows the split

**1,169,262** loans train the model, **5,691** are left out and **141,725** are tested.

# Boosted trees forecast best; tuning did not clearly beat fixed settings

Every forecaster on the same loans and test years; tuned rows are exploratory; August is a separate holdout

| Model | 2021 | 2022 | 2023 | 2024 | 2025 | ROC AUC | Settings | Used on |
|---|---|---|---|---|---|---|---|---|
| **Lookup and median baselines** | | | | | | | | |
| **Median only** | 10.51 | 10.44 | 10.67 | 10.04 | 7.38 | 0.5 | No settings | A3 |
| **Historical lookup** | 9.62 | 9.56 | 9.48 | 9.08 | 6.86 | 0.652–0.737 | No settings | Slide 17 |
| **Linear** 1 | | | | | | | | |
| **Linear, loan details** | 8.25 | 8.57 | 8.18 | 7.52 | 6.41 | 0.824–0.893 | Fixed, α 10 | Slides 14–15 |
| **Linear, with text** | 7.96 | 8.20 | 7.93 | 7.13 | 6.15 | 0.841–0.889 | Fixed, α 10 | Slides 14–15 |
| **Boosted trees, fixed settings** | | | | | | | | |
| **Boosted tree, loan details** | 7.50 | 6.96 | 6.72 | 5.70 | 4.87 | 0.870–0.909 | Fixed in advance | Slides 15, 17 |
| **Larger boosted tree** 2 | 8.05 | 6.61 | 6.53 | 5.34 | 4.80 | 0.848–0.906 | Fixed in advance | Slides 14–15 |
| **Larger tree, full text** | 7.89 | **6.55** | 6.51 | 5.35 | 4.73 | 0.853–0.910 | Fixed in advance | Slide 14 |
| **Tuned trees, exploratory** | | | | | | | | |
| **Tuned tree, loan details** 3 | 8.04 | 6.99 | 6.55 | **5.34** | 4.82 | — | Tuned each year | Exploratory |
| **Tuned tree, full text** | 7.87 | 6.88 | 6.53 | 5.38 | 4.73 | — | Tuned each year | Exploratory |
| **Tuned XGBoost, loan details** 4 | 8.05 | 7.10 | **6.49** | 5.37 | 4.74 | — | Tuned each year | Exploratory |
| **Tuned XGBoost, full text** | 7.83 | 7.00 | 6.50 | 5.40 | 4.67 | — | Tuned each year | Exploratory |
| **Text only and programme text** | | | | | | | | |
| **Linear, text only** | 9.10 | 9.12 | 9.04 | 8.55 | 6.55 | 0.725–0.776 | Fixed, α 10 | Slide 14 |
| **Linear, programme text** | 8.23 | 8.74 | 8.47 | 7.51 | 6.44 | 0.838–0.894 | Fixed, α 10 | A3 |
| **Linear, programme and text** | 8.05 | 8.39 | 8.12 | 7.23 | 6.17 | 0.847–0.889 | Fixed, α 10 | A3 |
| **Boosted tree, programme text** | 7.36 | 6.74 | 6.63 | 5.51 | 4.68 | 0.872–0.912 | Fixed in advance | A3 |
| **Tree, programme and text** | **7.34** | 6.76 | 6.61 | 5.53 | **4.65** | 0.873–0.912 | Fixed in advance | A3 |
| **August boosted trees** | 5.2031 days on the 2024–25 holdout; $R^2$ 0.5420 | | | | | 0.8997 | Library defaults | A3 |

Lowest error that year  Close to the year's lowest  Further from it

**Figure A3.** Shading: gap to each year's lowest. Tuned: 24 settings per family on the prior year, chosen on loan details, reused with text. Programme text: a proxy.

### 1 Loan details do the most work

Moving from a lookup to a model on loan details cuts error in **every year**.

### 2 Bigger is not always better

The larger tree beats the benchmark tree in 4 of 5 years, but **not in 2021**.

### 3 Tuning did not clearly help

On loan details, tuned trees were lowest in 3 of 5 years, **by at most 0.06 days**.

### 4 XGBoost is no clear winner

It beats the tuned boosted tree in **2 of 5 years**; both are exploratory.

**How to read the scores**

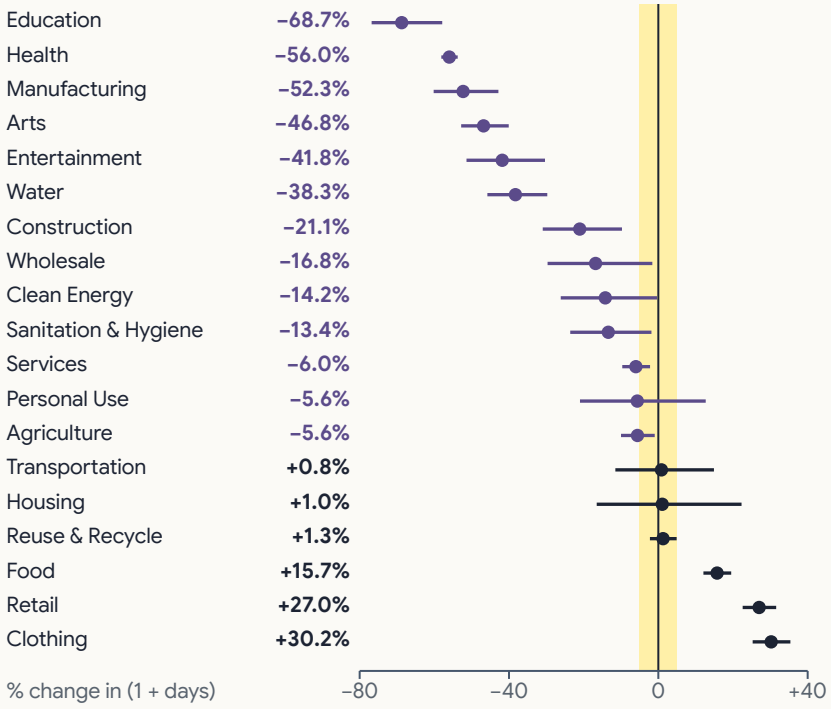
**MAE**: average miss in days; lower is better.
**$R^2$**: variation explained; below 0 is worse than the mean.
**ROC AUC**: ranks same-day loans; 0.5 is random, 1 is perfect.
**Brier** 0.1156: same-day probability error; lower is better.

# Loan purpose tracks funding speed; sanitation's edge is mostly context

Programme-adjusted model on all completed loans; associations, not effects of changing a loan

### Sectors against the average loan

3 1




### Activities with 1,000+ loans, against farming

| Fastest | Loans | |
|---|---|---|
| Education provider | 6,095 | −73.0% |
| Higher education costs | 17,561 | −68.7% |
| Primary/secondary school costs | 16,034 | −60.7% |
| Metal Shop | 1,209 | −54.2% |
| Pharmacy | 2,222 | −53.5% |
| **Slowest** | **Loans** | |
| Cosmetics Sales | 5,276 | +61.5% |
| Pub | 1,695 | +49.5% |
| Plastics Sales | 1,155 | +43.5% |
| Beauty Salon | 10,896 | +42.4% |
| Charcoal Sales | 13,565 | +41.7% |

**Toilets and sanitation** 2

Raw mean **0.9 days** among activities with 1,000+ loans.

Activity against farming: **−5.7%** [−19.7%, +10.7%], n 6,715; range includes zero.

Sector against the average loan: **−13.4%** [−23.6%, −1.8%]; same-day **+8.8 pp** [+6.3, +11.2], n 6,905.

Sample themes: 1.5 to 13.5 days raw; adjusted on A5.

● Faster than average (negative) ● Slower than average (positive)

**Figure A4.** Change in (1 + days), country-clustered 95% ranges. Sector: against the average loan. Activity: against Farming, with 1,000+ loans.

### 1 Purpose tracks funding speed

Education runs **−68.7%** and clothing **+30.2%** against the average loan.

### 2 Sanitation is fast mostly through context

Fast in raw means; with controls the sector is **modestly faster** and the activity is **not distinct from farming**.

### 3 These are not levers

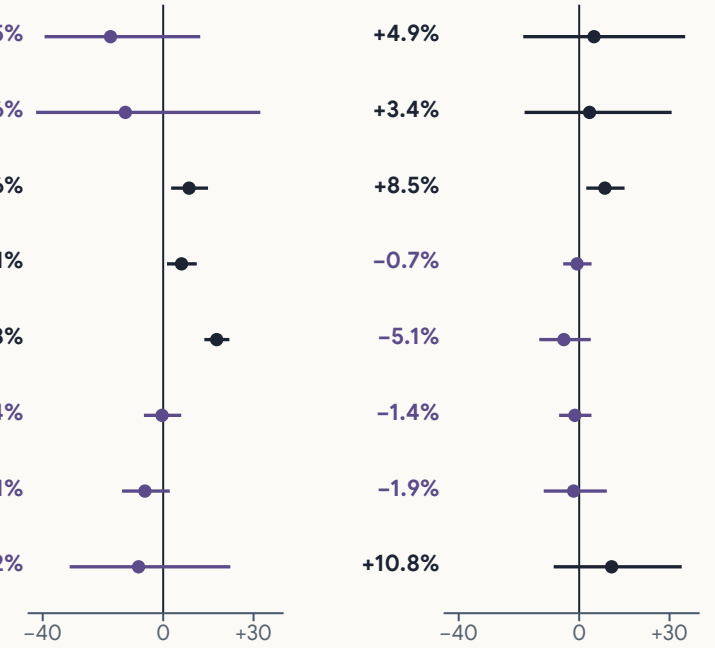
Associations among completed loans; a loan's **purpose is not something to change**.

# Sanitary and water themes look fast, but are not distinct after controls

Exploratory NMF themes from a 20,000-description sample; associations among completed loans

**Eight themes: raw speed against adjusted contrasts**

| Theme and top words | Loans | Mean | Median | 1 Adjusted for loan details, country, time | 2 Also adjusted for activity |
|---|---|---|---|---|---|
| **Sanitary and toilets**<br>sanitary, toilet, risks, hazard, health | 1,029 | 1.5 | 0.5 | −17.5% | +4.9% |
| **Water and filters**<br>water, village, filter, clean, drinking | 673 | 1.8 | 0.3 | −12.6% | +3.4% |
| **Pig raising**<br>pigs, pig, raise, supplies, nwtf | 1,590 | 7.3 | 1.2 | +8.6% | +8.5% |
| **Philippines business**<br>business, nwtf, philippines, php, entrepreneur | 2,361 | 7.8 | 1.5 | +6.1% | −0.7% |
| **General store**<br>store, general, items, canned, goods | 1,839 | 9.0 | 1.8 | +17.8% | −5.1% |
| **Business and family**<br>business, loan, income, children, years | 8,883 | 10.7 | 4.4 | −0.4% | −1.4% |
| **Farm inputs**<br>farm, farming, inputs, quality, farmer | 2,739 | 12.5 | 6.0 | −6.1% | −1.9% |
| **Group solar and farms**<br>group, solar, acre, total, fund | 886 | 13.5 | 10.1 | −8.2% | +10.8% |




● Faster than average (negative) ● Slower than average (positive)

**Figure A5.** NMF on a 20,000-description sample, 43 countries. Change in (1 + days) against the average theme, country-clustered 95% ranges, not corrected for eight comparisons. Days: raw mean and median. Sample only.

### 1 Two themes look very fast

Sanitary and water themes average **1.5 and 1.8 days**; the other six, 7.3–13.5.

### 2 That speed is mostly context

They **fund fast, but that speed largely reflects which countries and loan types they are**.

### 3 Only pig raising holds up

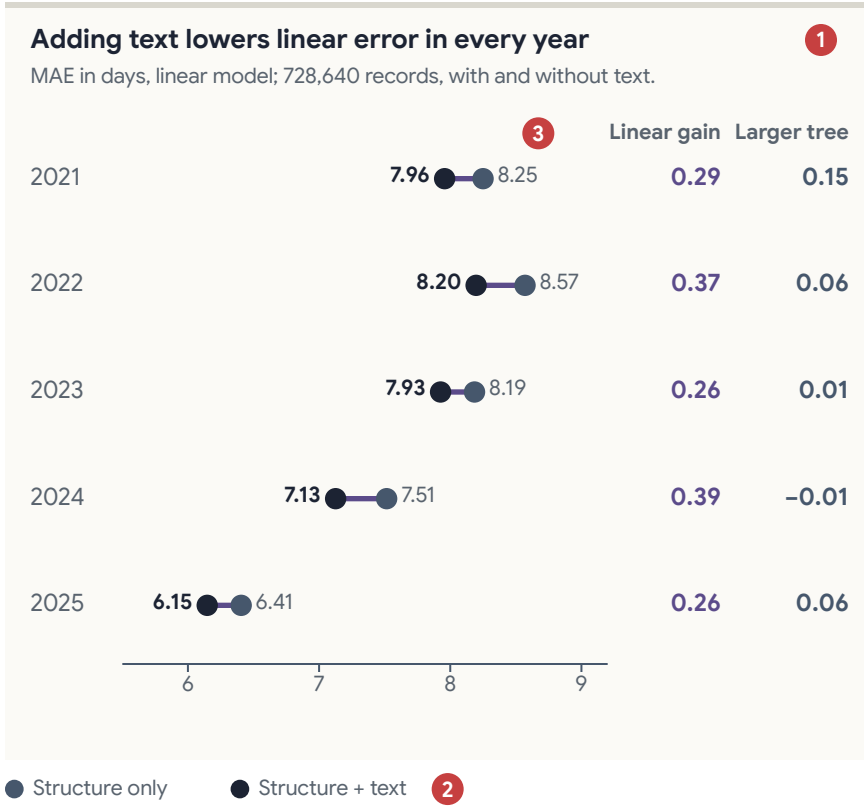
Three themes are slower with loan details held fixed; with activity too, only **pig raising** is, at +8.5%.

### 4 A sample, not a lever

One sample, associations only; a loan's **theme is not something to change**.

# Text lowers a linear model's error; a stronger tree barely changes

Forecast benchmark: linear model, structure only versus structure plus text, identical evaluation rows



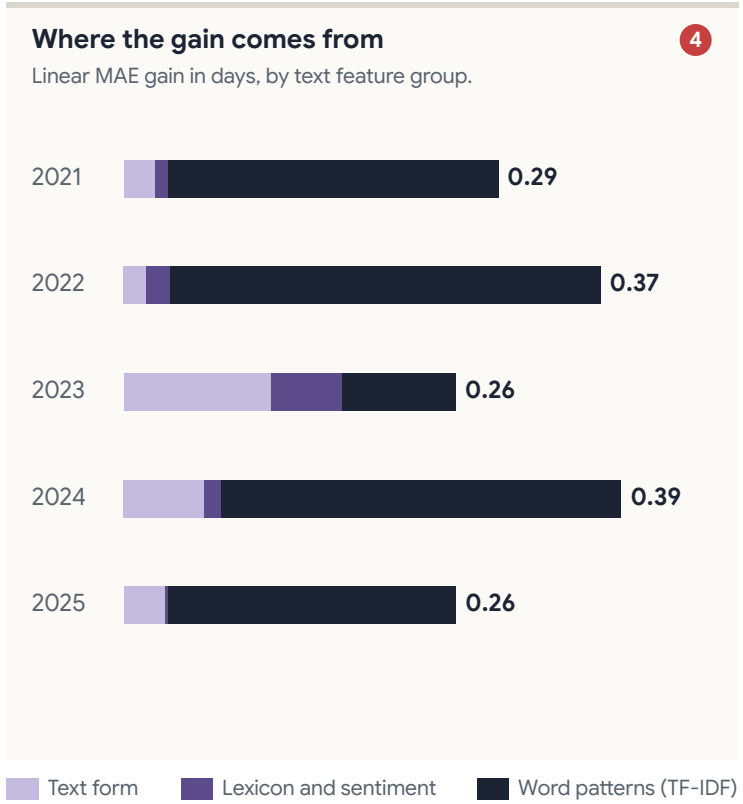




**Figure A6.** Linear-model MAE in days (left) and its reduction by text feature group (right), identical evaluation rows. Larger tree: full-text gain, fixed settings.

[1] Groups added in order: form, then lexicon and sentiment, then TF-IDF; another order would split the gain differently.

### 1 A small linear gain

Gains of **0.26–0.39 days** on 728,640 evaluation records; about half is context.

### 2 Forecasting is not persuasion

Text may encode **purpose, place or programme templates**, not only persuasion.

### 3 No single country drives it

Removing any single country **leaves the gain positive** in every year (exploratory check).

### 4 Word patterns carry most of it

TF-IDF gives the largest share in **4 of 5 years**; lexicon and sentiment add only **0.003–0.055 days**.

# Text became numbers through classical NLP, not a large language model

Every text feature in the deck, how it was built, and where its limits are

| Technique | Settings | Where used | Limit |
|---|---|---|---|
| **Description form** 1<br>Length and structure of the text | **Characters, words, sentences**, average word and sentence length | Explanatory models and forecasts | Says nothing about content |
| **Word-list counts** 3<br>Eight framing measures, including first and third person | Theory-guided lists of **4–13 entries**; **rate per 100 words** | Explanatory, robustness and equivalence; forecasts | Not yet validated (gate 1); counts words, not meaning |
| **VADER sentiment**<br>Positive or negative tone | **Rule-based lexicon**; compound score from **–1 to 1** | Explanatory, robustness and equivalence; forecasts | General-purpose lexicon, not built for loan text |
| **TF-IDF word patterns** 2<br>Distinctive words and two-word phrases | Up to **5,000 features**, in at least 5 training loans; fitted on **training years only** | Linear forecast | May pick up place and programme templates |
| **Compressed text features**<br>TF-IDF reduced for the boosted tree | Truncated SVD, **64 components**; fitted on **training years only** | Larger boosted tree | Hard to interpret |
| **NMF topics** 4<br>Eight story themes | TF-IDF up to **1,000 features**; **20,000-loan sample** | Appendix A5 only | Exploratory; sample only, not full-data evidence |

Fixed rules  Fitted on training years only  Exploratory, sample only

**Figure A7.** Methods as implemented in src/ and the exploratory notebook. No large language model or pretrained embeddings were used.

### 1 Rules need no training

Form, word lists and VADER are **fixed rules**, so nothing is learned from the outcome.

### 2 Learned features stay on training years

**TF-IDF** and its compressed form are fitted on training years only.

### 3 Classical NLP, not yet validated

No large language model; the **word lists still need validating**.

### 4 Themes stay a sample view

NMF themes come from a **20,000-loan sample**, so they appear only on A5.

# Every result follows one path from raw loans to tested claims

One cleaned dataset feeds four analyses; each lane ends at the slides that use it

**Raw data** (1)
1,453,846 loan records, Kiva listings 2016–2025, completed durations only

→ **Cleaning**
6 negative durations removed, HTML and extra spaces stripped: 1,453,840 eligible loans

→ **Slides 4–6, 9**

| Analysis | Features | Models | Tests (3) | In the deck |
|---|---|---|---|---|
| **Explanatory models** | ● Loan details<br>● Description form<br>● Word lists<br>● VADER sentiment | Regressions on all eligible loans | HC3 and country-clustered errors; wild cluster bootstrap; ±5% equivalence test | **Slides 10–13, A1** |
| **Forecast benchmark** | ● The features above<br>● TF-IDF<br>● Compressed TF-IDF | Linear forecasts, boosted trees, tuned trees | Month and country resampling; text placebo | **Slides 14–15, A3, A6** |
| **Slow-funding flag** | ● Loan details only | Boosted tree against a historical lookup | Lift, within-country precision, ranges for its margin | **Slide 17** |
| **Exploratory themes** | ● NMF topics, sample only | Adjusted contrasts by theme | Country-clustered 95% ranges | **A5** |

● Fixed rules ● Fitted on training years only (2) ● Exploratory, sample only

**Figure A8.** Features: slide 7 and A7. Later checks use the same eligible loans unless their slide says otherwise. Models train on earlier years and test on later ones. Loan photos were not analysed: fetching them at scale needs Kiva's permission. (4)

### (1) One population throughout

The analyses use **the same 1,453,840 eligible loans**, unless a slide says otherwise.

### (2) Nothing learned from the future

Learned text features are **fitted on training years only**.

### (3) Tests are layered

Each claim passes **several checks**, from clustered errors to a text placebo.

### (4) Photos stayed out

Loan photos were **not analysed**; fetching them at scale needs Kiva's permission.